

Q1.

(a) Solve $\frac{2x-1}{3} - \frac{5x-3}{6} = 2$

(b) Evaluate to 2 decimal places $e^x \sin 2x$ when $x = 2$.

(c) Write 7.21 as a mixed fraction.

(d) A person deposits \$630 into a savings account for 10 years at an interest rate of 4.35 % p.a.

Find the value of the account at the end of 10 years.

(e) Rationalise the denominator and simplify $\frac{7-\sqrt{2}}{3+\sqrt{2}}$.

Question 2 (Start A New Page)

(a) Differentiate y wrt x :

(i) $y = e^2 \pi$

(ii) $y = x \sec x$

(iii) $y = 2 \ln(x^3 + 4)$

(iv) $y = \tan^3 2x$

(b) Find the equation of the tangent to the curve $y = \frac{x}{x^2 + 1}$ at $x = 2$.

Write your answer in general form.

Question 3 (Start A New Page)

(a) Find the primitive function of $\frac{e^{3x}}{3} - \cos 2x$.

(b) Find:

(i) $\int (7x-3)^2 dx$

(ii) $\int \sqrt{x} (2x+1) dx$

(iii) $\int \frac{e^{2x} + 1}{e^x} dx$

(c) Evaluate (i) $\int_0^{\frac{\pi}{4}} 4 \sin 2x dx$

(ii) $\int_0^3 \frac{2x}{4x^2 + 1} dx$

Question 4 (Start A New Page).

(a) A surveyor records the following diagram.

$\angle ABC = 125^\circ$, $\angle ABD = 47^\circ$, $\angle BAD = 110^\circ$, $\angle BAC = 41^\circ$

Find the lengths (2 decimal places) of:

(i) BD

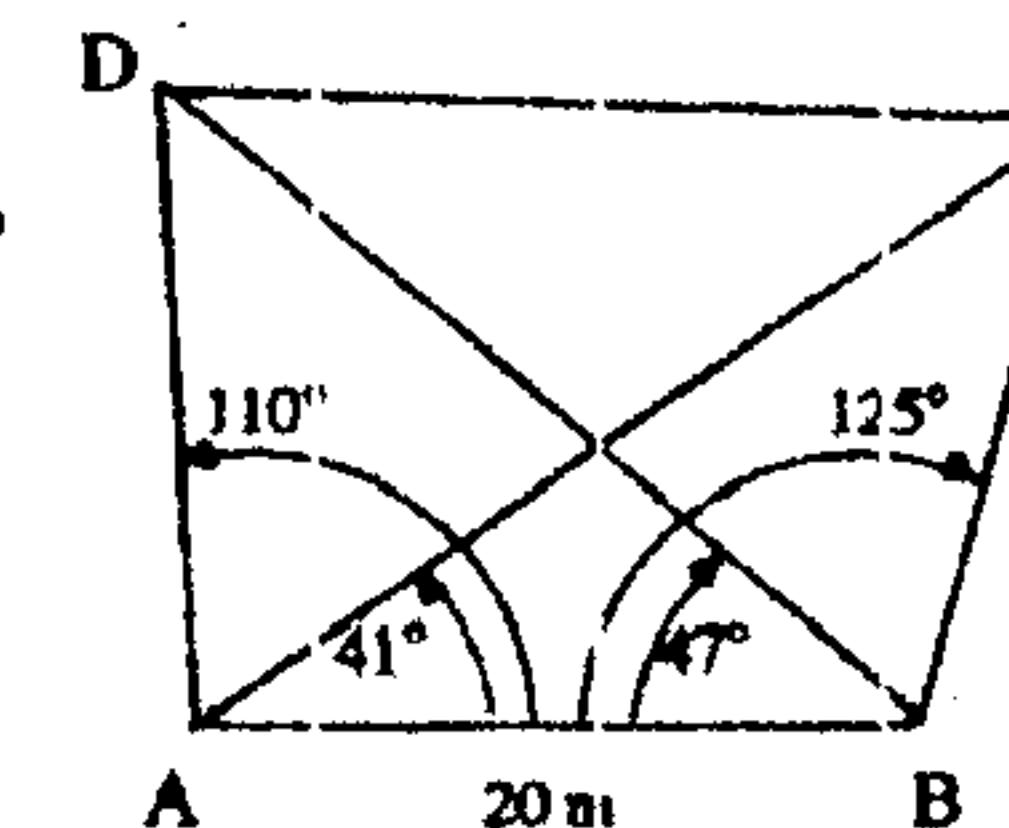
(ii) CD if $BC = 54$ metres.

(b) Given $A(-5, 4)$, $B(1, 7)$, $C(9, 1)$ and $D(-1, -4)$

(i) Show (a) ABCD is a trapezium

(b) $AD \perp CD$

(ii) Find the area of AECD



Question 5 (Start A New Page)

(a) Graph $y = \frac{x+3}{x-2}$

(b) Simplify $1 + \frac{5}{x-2}$

(c) The region bounded by the curve $y = \frac{x+3}{x-2}$ and the lines $x = 3$, $x = 4$, $y = 0$, is rotated around the x axis. Find the volume of revolution.

(d) Find the area bounded by the curve $y = \sqrt{9-x^2}$ and the positive x and y axes.

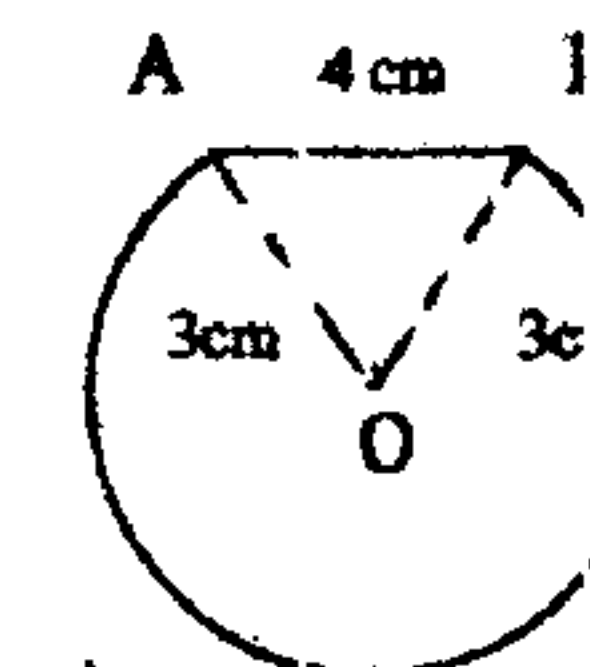
Question 6 (Start A New Page)

(a) The cross sectional area of a circular cam centre O is shown.

Show that $\angle AOB = 1.459$ radians.

hence find (i) the perimeter of the cam (to 1 decimal place)

(ii) the cross sectional area of the cam (to 1 decimal place).



(b) A compound substance decomposes into its elements at a rate proportional to the amount of compound present, ie $\frac{dM}{dt} = -kM$ where M is the mass of the compound present after t seconds and k is the proportionality constant.

(i) Show $M = Ae^{-kt}$ is a solution of the differential equation where A is a constant.

(ii) If 900 grams decomposes to 350 grams in 50 seconds

(a) show $A = 900$ and $k = \frac{1}{50} \ln\left(\frac{18}{17}\right)$,

(b) find the amount compound (nearest gram) remaining after 140 seconds.

Question 7 (Start A New Page).

Given the function $y = \frac{10}{3 + 2 \sin x}$ in the domain $0 \leq x \leq 2\pi$

- (a) find the location of all the stationary points and determine their nature
- (b) graph the function in the given domain
- (c) using 5 function values evaluate by Simpson's rule $\int_0^{2\pi} \frac{10 \, dx}{3 + 2 \sin x}$

Question 8 (Start A New Page).

- (a) A man borrows \$ 13500 at 9.9% p.a. reducible monthly interest for 4 years, and agrees to repay the loan in equal monthly instalments. Find the value of the monthly repayment.
- (b) Twelve students are to be chosen from twenty students of equal ability.

Find the probability that :

- (i) three particular students A , B , C are chosen
- (ii) students A and B are chosen but student C is not chosen
- (iii) none out of students A , B , or C are chosen
- (iv) at least one of the students A , B , or C are chosen.

Question 9 (Start A New Page)

- (a) Find the values of k to make $y = (k + 2) x^2 + 4\sqrt{3} x + 5 - k$ positive definite.
- (b) Given the function $g(x) = \frac{e^x - e^{-x}}{2}$,
 - (i) show that $g(x)$ is an increasing function for all values of x
 - (ii) find the minimum value of the gradient of $g(x)$ and where this occurs.
 - (iii) graph $y = g(x)$

Question 10 (Start A New Page).

- (a) A particle moves with velocity v m/s in time t seconds according to :

$$v = \frac{6}{\sqrt{t+1}}$$

Find (i) the acceleration as a function of time t

- (ii) the displacement x as a function of time t if initially the particle was 2 metres to the right of the origin.

- (b) In a pentagon ABCDE , AB = AE , BC = ED and BD = EC.

Prove that AC = AD

QUESTION 1

- (a) $x = -11$
 (b) -5.59
 (c) $7^{7/33}$
 (d) $\$995.03$
 (e) $\frac{23-10\sqrt{2}}{7}$

QUESTION 2

- (a)(i) 0
 (ii) $x \sec x \tan x + \sec x$
 (iii) $\frac{6x^2}{x^3+4}$
 (iv) $6 \tan^2 2x \cdot \sec^2 2x$
 (b) $3x + 25y - 16 = 0$

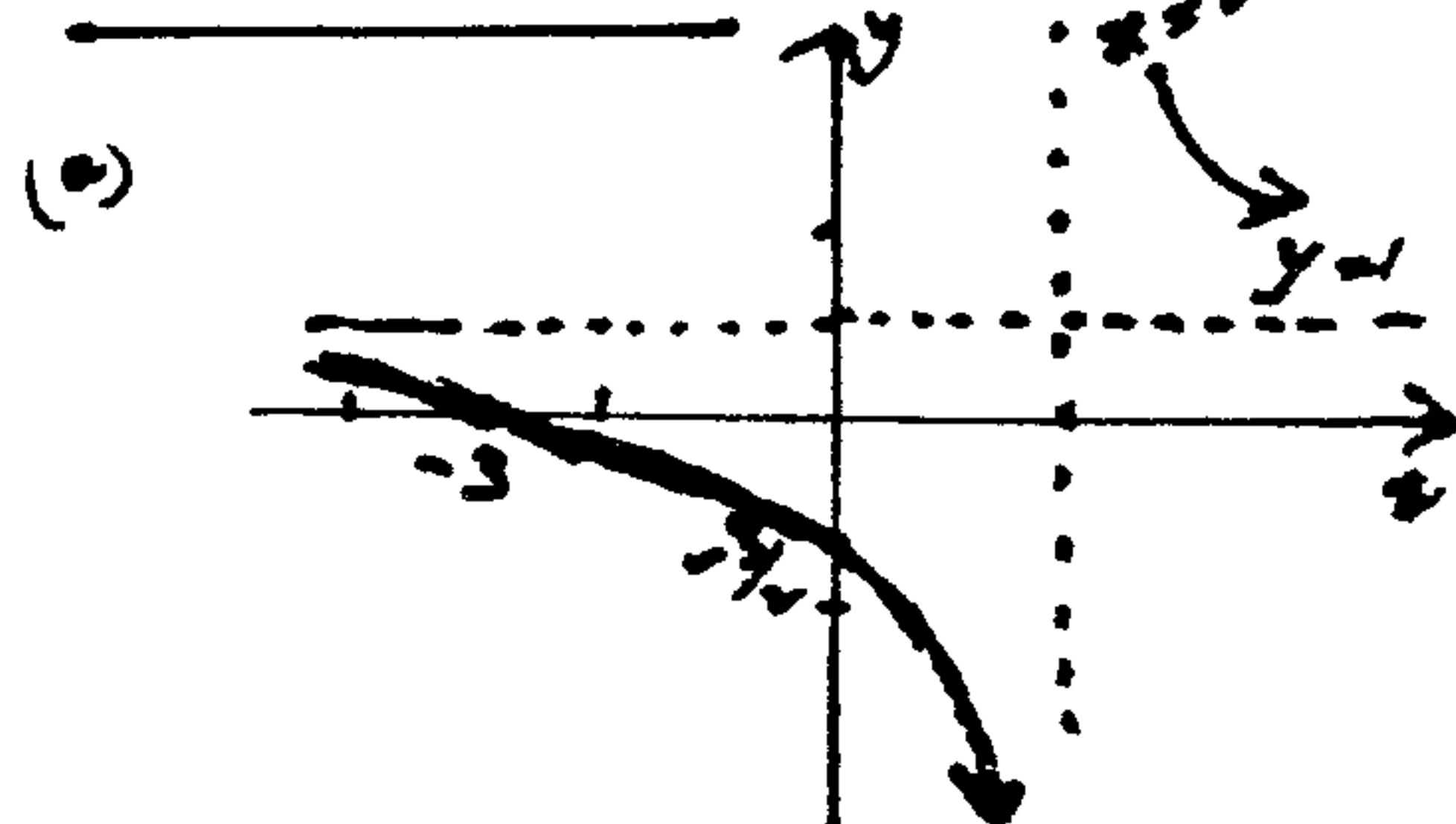
QUESTION 3

- (a) $\frac{1}{4}e^{3x} - \frac{1}{2}\sin 2x + c$
 (b)(i) $\frac{1}{21}(7x-3)^3 + c$
 (ii) $\frac{4}{5}x^{5/2} + \frac{2}{3}x^{3/2} + c$
 (iii) $e^x - e^{-x} + c$
 (c)(i) 2
 (ii) $\frac{1}{4} \ln 37$

QUESTION 4.

- (a)(i) 48.10 m
 (ii) 64.42 m
 (b)(i) $AB \parallel CD$ ($m = \frac{1}{2}$)
 (ii) $m(AD) \cdot m(CD) = -1$
 (iii) $80 u^2$

QUESTION 5



(b) $\frac{x+3}{x-2}$

- (c) $\frac{\pi}{2}(27 + 20 \ln 2)$
 (d) $\frac{9\pi}{4} u^2$

QUESTION 6

- (a)(i) 18.5 cm
 (ii) 26.2 cm^2
 (b)(a)(i) -
 (ii) -
 (b) 767 g .

QUESTION 7

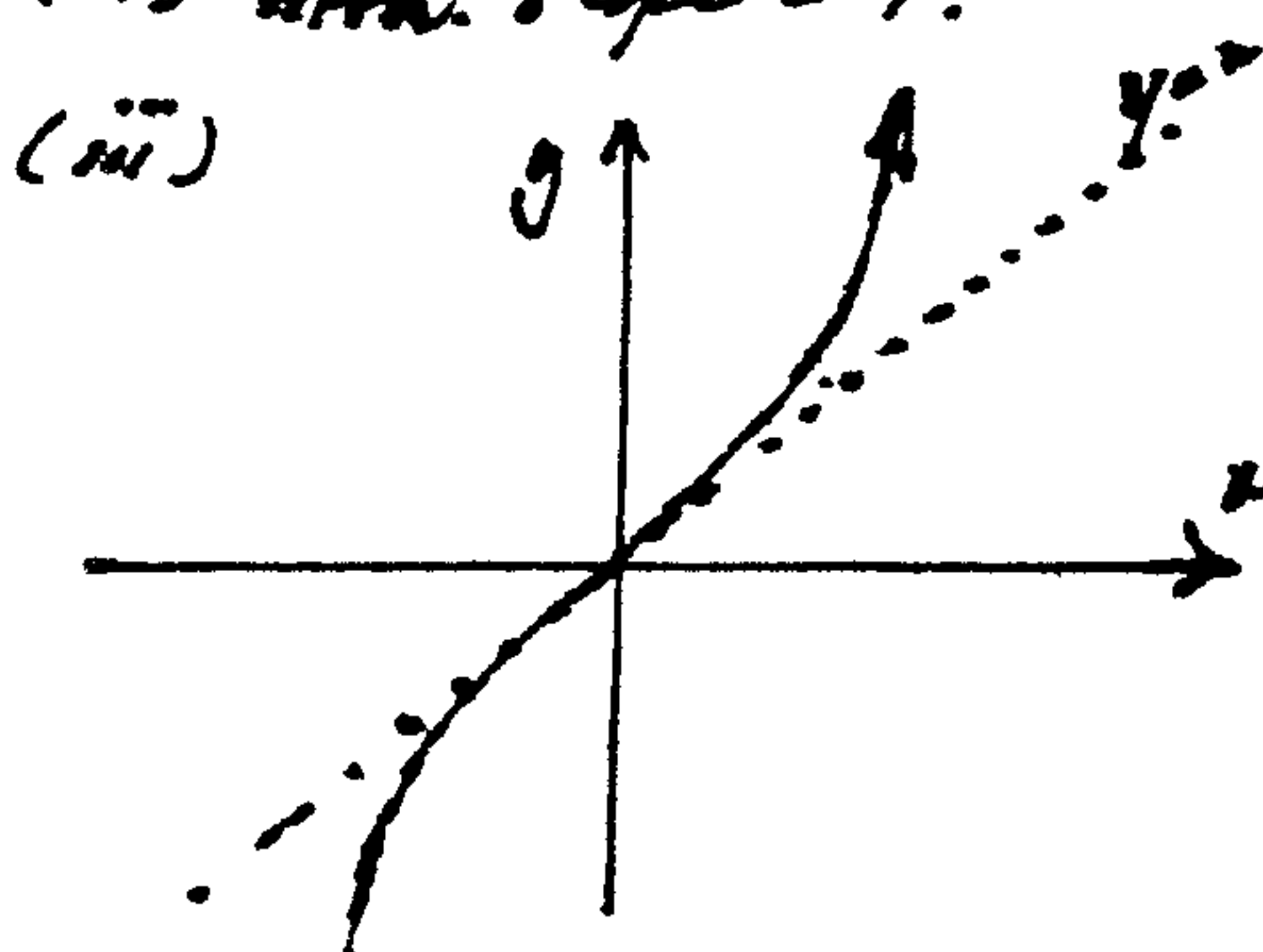
- (a) min tp $(\pi/2, 2)$
 max tp $(3\pi/2, 10)$
 (b) -
 (c) $\frac{92\pi}{9}$

QUESTION 8

- (a) $\$341.75$
 (b)(i) $11/57$
 (ii) $44/285$
 (iii) $14/285$
 (iv) $271/285$

QUESTION 9

- (a) $1 < k < 2$
 (b)(i) $g'(x) = \frac{1}{2}(e^x + e^{-x})$
 so for all x
 (ii) min. slope = 1.
 (iii)



QUESTION 10

- (a)(i) $a = \frac{-3}{(t+1)^{3/2}}$
 (ii) $x = 12\sqrt{t+1} - 10$
 (b) -